Module 3: Syntax Analysis

- By Tina D’abreo

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Part-Of-Speech tagging (POS)

- Following are the Parts-of-Speech:
  - Nouns:
    - Proper Nouns – Ravi, IBM, India
    - Common Nouns – Child, Dog, House
  - Pronoun:
  - Verbs
  - Adverbs
  - Preposition:
    - in, out, besides, in front of, below, opposite, etc.
  - Adjective
  - Conjunctions:
    - and, or, for, yet, although, because, not only, etc.
  - Interjections:
    - oh, wow, alas, yippee, etc. It is always followed by an exclamation mark.
  - Particle
  - Determiners:
    - Subtype → the articles : a, an, the
    - This, That

Part-Of-Speech tagging (POS)

- In grammar, a **part-of-speech (POS)** is known as ***word class or grammatical category,*** which is a category of words that have similar grammatical properties.
- The English language has four major word classes:
  - Nouns, Verbs, Adjectives, and Adverbs.
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Part-Of-Speech tagging (POS)

- Parts-of-speech can be divided into two broad supercategories:
  - Closed class types
    - Closed classes are those with a relatively fixed/number of words, and we rarely add new words to these POS, such as prepositions.
    - Closed class words are generally functional words like of, it, and, or you, which tend to be very short, occur frequently, and often have structuring uses in grammar.
    - Example of closed class-
      - Determiners: a, an, the
      - Pronouns: she, he, I, others
      - Prepositions: on, under, over, near, by, at, from, to, with

Part-Of-Speech tagging (POS)

- ➔ Parts-of-speech can be divided into two broad supercategories:
  - ◆ Open class types
    - Open Classes are mostly content-bearing, i.e., they refer to objects, actions, and features; it's called open classes since new words are added all the time.
    - By contrast, nouns and verbs, adjectives, and adverbs belong to open classes; new nouns and verbs like iPhone or to fax are continually being created or borrowed.
    - Example of open class-
      - Nouns: computer, board, peace, school
      - Verbs: say, walk, run, belong
      - Adjectives: clean, quick, rapid, enormous
      - Adverbs: quickly, softly, enormously, cheerfully

Part-Of-Speech tagging (POS)

- ➔ Part-of-Speech (POS) tagging is a natural language processing technique that involves assigning specific grammatical categories or labels (such as nouns, verbs, adjectives, adverbs, pronouns, etc.) to individual words within a sentence.
- ➔ Example: “Our dog chased a brown cat away from the home”

Word	**POS Category**
Our	Pronoun
Dog, cat, home	Noun
A, the	Determiners
Brown	Adjective
Away	Adverb
From	Preposition

Part-Of-Speech tagging (POS)

- ➔ Part-of-Speech (POS) tagging problem is defined as a problem to identify the actual category for each of the word.
- ➔ Given a text in english, identify the parts of speech of each word in the sentence
  - ◆ “The boy put the toys in the bag”

POS Category	Word
Noun (N)	Boy, toys, bag
Determiners (Det)	the
Verb (V)	put
Preposition (P)	in

Part-Of-Speech tagging (POS)

- ➔ Part-of-speech tagging (or just tagging for short) is the process of assigning a part-of-speech or other syntactic class marker to each word in a corpus.
- ➔ Because tags are generally also applied to punctuation, tagging requires that the punctuation marks (period, comma, etc) be separated off of the words.
- ➔ Thus tokenization o is usually performed before, or as part of, the tagging process, separating commas, quotation marks, etc., from words, and disambiguating end-of-sentence punctuation

Part-Of-Speech tagging (POS)

- ➔ The input to tagging algorithm is a string of words and a specific tagset. The output is a single best tag for each word.
- ➔ Example:
  - ◆ Book/VB that/DT flight/NN
  - ◆ Does/VBZ that/DT flight/NN serve/VB dinner/NN ?/.

Difficulties /Challenges in POS tagging

- ➔ Tagging decisions are difficult to make.
- ➔ Automatically assigning tags to the words is not trivial
- ➔ For example, book is ambiguous
- ➔ That is, it has more than one possible usage and part-of-speech.
- ➔ It can be a verb (***as in book that flight or to book the suspect***) or a noun (***as in hand me that book, or a book of matches***).
- ➔ Similarly ***that*** can be a determiner (as in Does that flight serve dinner), or a complementizer (as in I thought that your flight was earlier).
- ➔ The problem of POS-tagging is to resolve these ambiguities, choosing the proper tag for the context.

Difficulties /Challenges in POS tagging

- ➔ Most words in English are unambiguous; i.e., they have only a single tag.
- ➔ But many of the most common words of English are ambiguous
  - ◆ For example ‘can’ can be an auxiliary (‘to be able’),
  - ◆ A noun (‘a metal container’), or
  - ◆ A verb (‘to put something in such a metal container’)).

Difficulties /Challenges in POS tagging

		Original 87-tag corpus		Treebank 45-tag corpus	
Unambiguous (1 tag)		**44,019**		**38,857**	
Ambiguous (2–7 tags)		**5,490**		**8844**	
Details:	2 tags	4,967		6,731	
	3 tags	411		1621	
	4 tags	91		357	
	5 tags	17		90	
	6 tags	2	(*well*, *beat*)	32	
	7 tags	2	(*still*, *down*)	6	(*well*, *set*, *round*, *open*, *fit*, *down*)
	8 tags			4	(*'s*, *half*, *back*, *a*)
	9 tags			3	(*that*, *more*, *in*)

Figure 5.10 The amount of tag ambiguity for word types in the Brown corpus, from the ICAME release of the original (87-tag) tagging and the Treebank-3 (45-tag) tagging. Numbers are not strictly comparable because only the Treebank segments *'s*. An earlier estimate of some of these numbers is reported in DeRose (1988).

Tag set for English

- ➔ There are small number of popular tagsets for English, many of which evolved from the 87-tag tagset used for the Brown corpus
- ➔ Brown Corpus is a 1 million word collection of samples from 500 written texts from different genres which was assembled at Brown University in 1963-64.
- ➔ This corpus was tagged with parts-of-speech by first applying the TAGGIT program and then hand-correcting the tags. → **Brown Corpus Tagset**
- ➔ Besides this original Brown tagset, two of the most commonly used tagsets are:
  - ◆ The small **45-tag Penn Treebank tagset** (Marcus et al., 1993),
  - ◆ The medium-sized **61 tag C5 tagset** used by the Lancaster UCREL project's CLAWS (the Constituent Likelihood Automatic Word-tagging System) tagger to tag the British National Corpus

Tag set for English

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	*and, but, or*	SYM	Symbol	+,%, &
CD	Cardinal number	*one, two, three*	TO	"to"	*to*
DT	Determiner	*a, the*	UH	Interjection	*ah, oops*
EX	Existential 'there'	*there*	VB	Verb, base form	*eat*
FW	Foreign word	*mea culpa*	VBD	Verb, past tense	*ate*
IN	Preposition/sub-conj	*of, in, by*	VBG	Verb, gerund	*eating*
JJ	Adjective	*yellow*	VBN	Verb, past participle	*eaten*
JJR	Adj., comparative	*bigger*	VBP	Verb, non-3sg pres	*eat*
JJS	Adj., superlative	*wildest*	VBZ	Verb, 3sg pres	*eats*
LS	List item marker	*1, 2, One*	WDT	Wh-determiner	*which, that*
MD	Modal	*can, should*	WP	Wh-pronoun	*what, who*
NN	Noun, sing. or mass	*llama*	WP$	Possessive wh-	*whose*
NNS	Noun, plural	*llamas*	WRB	Wh-adverb	*how, where*
NNP	Proper noun, singular	*IBM*	$	Dollar sign	*$*
NNPS	Proper noun, plural	*Carolinas*	#	Pound sign	#
PDT	Predeterminer	*all, both*	"	Left quote	' or "
POS	Possessive ending	*'s*	"	Right quote	' or "
PRP	Personal pronoun	*I, you, he*	(	Left parenthesis	[, (, {, <
PRP$	Possessive pronoun	*your, one's*	)	Right parenthesis	],), }, >
RB	Adverb	*quickly, never*	,	Comma	,
RBR	Adverb, comparative	*faster*	.	Sentence-final punc	. ! ?
RBS	Adverb, superlative	*fastest*	:	Mid-sentence punc	: ; ... – -
RP	Particle	*up, off*			

Figure 5.6 Penn Treebank part-of-speech tags (including punctuation).

Tag set for English

➔ Noun Types

POS Type	Explanation	Examples
NN	Singular Common Noun	Woman, Orange, Table
NNS	Plural Common Noun	Women, Oranges, Tables
NNP	Singular Proper Noun	Priya, Zenith, Jack
NNPS	Plural Proper Noun	Indians, Americas

Tag set for English

➔ Verb Types

POS Type	Explanation	Examples
VB	Base form of a Verb	walk, play, eat, read
VBD	Past Tense of a Verb	Walked, played, ate, read
VBN	Past Participle of a Verb	Walked, played, eaten, read
VBG	Gerund form of a Verb	fishing, walking, reading
VBZ	3rd person verb on present tense	Walks, plays, eats, read, is
VBP	Non 3rd person verb on present tense	Walk, play, eat, read, am, are
MD	Modal Verb	Can, may, should

Tag set for English

➔ Adjective and Adverb Types

POS Type	Explanation	Examples
JJ	Adjective	Intelligent, small, fast
JJR	Comparative Adjective	Better, smaller, faster
JJS	Superlative Adjective	Best, smallest, fastest
RB	Adverb	Back, behind, fast, slow
RBR	Comparative Adverb	Slower, faster
RBS	Superlative Adverb	Slowest, fastest

Tag set for English

➔ Pronoun, Determiner, Preposition Types

POS Type	Explanation	Examples
PRP	Pronoun	He, she, they, I, we
PRP$	Possessive Pronoun	His, her, your, our
POS	Possessive Marker	India's, Asian's
DT	Determiner	The, a
CC	Coordinating Conjunction	And, or, also, but
IN	Preposition	In, under, of, from, with
CD	Cardinal Number	20, two

Tag set for English

Statement: **I am a girl.**

Word	POS Type	Explanation
I	PRP	Pronoun
am	VBP	Non 3rd person verb on present tense
a	DT	Determiner
girl	NN	Singular Common Noun

Statement: **Kavya is a intelligent girl.**

Word	POS Type	Explanation
Kavya	NNP	Singular Proper Noun
is	VBZ (Aux. verb)	3rd person verb on present tense
a	DT	Determiner
intelligent	JJ	Adjective
girl	NN	Singular Common Noun

Tag set for English

Statement: **She plays tennis.**

Word	POS Type	Explanation
She	PRP	Pronoun
plays	VBZ	3rd person verb on present tense
tennis	NN	Singular Common Noun

Statement: **They play football.**

Word	POS Type	Explanation
They	PRP	Pronoun
play	VBP	Non 3rd person verb on present tense
football	NN	Singular Common Noun

Tag set for English

Statement: **And now for something completely different**

Word	POS Type	Explanation
And	CC	Coordinating Conjunction
now	RB	Adverb
for	IN	Preposition
Something	NN	Singular Common Noun
completely	RB	Adverb
different	JJ	Adjective

Tag set for English

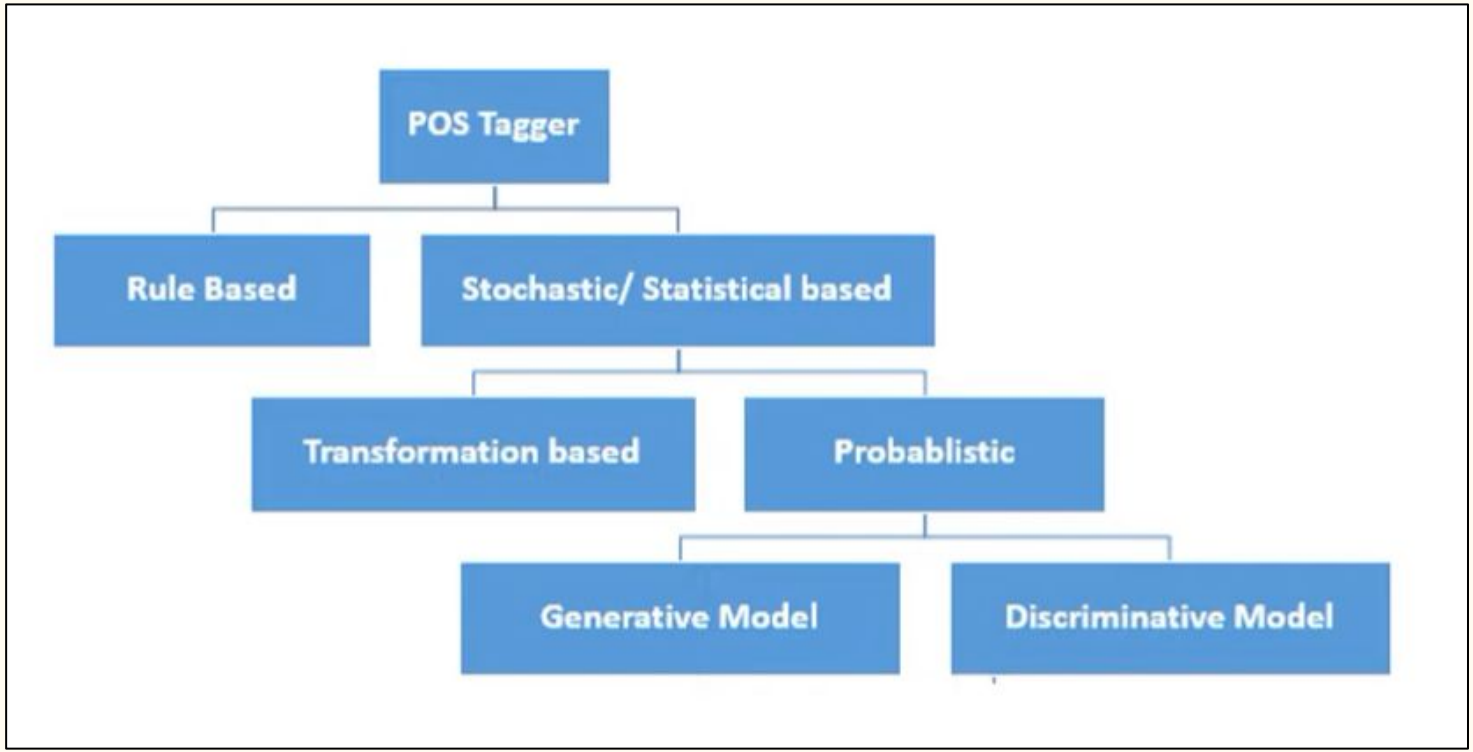
Statement: **They refuse to permit us.**

They	refuse	to	permit	Us
PRP	VBP	Det	VB	PRP

Statement: **They refuse to permit us to obtain refuse permit.**

They	refuse	to	permit	us	To	obtain	refuse	permit
PRP	VBP	Det	VB	PRP	Det	VB	NN	NN

POS Taggers


POS Tagger
Rule Based
Stochastic/ Statistical based
Transformation based
Probablistic
Generative Model
Discriminative Model

POS Taggers - Rule based Tagger

- ➔ Rule-based part-of-speech tagging is the oldest approach that uses hand written rules for tagging
- ➔ Rule-based taggers depends on dictionary or lexicon to get possible tags for each word to be tagged.
- ➔ This algorithm automatically assigns part-of-speech that were based on a two stage architecture (Harris, 1962; Klein and Simmons, 1963; Greene and Rubin, 1971).
  - ◆ The ***first stage*** used a dictionary to assign each word a list of potential parts-of-speech.
  - ◆ The ***second stage*** used large lists of hand-written disambiguation rules to winnow down this list to a single part-of-speech for each word.
- ➔ Example:
  - ◆ I want to **play/VB** a sport on weekend
  - ◆ I want to perform a **play/?**
    - *Rule: An ambiguous word is a Noun rather than a verb if it follows a determiner*

POS Taggers - Rule based Tagger

- ➔ The rules may be context-pattern rules or as regular expressions compiled into finite-automata that are intersected with lexically ambiguous sentence representation
- ➔ TAGGIT is the first large rule-based tagger that used context-pattern rules
- ➔ Limitations:
  - ◆ Hard coded rules are required
  - ◆ Rules has to be updated on language change
  - ◆ Lexicon needs to be checked every time which is not efficient approach
  - ◆ Strong language expert team is required

POS Taggers - Rule based Tagger

➔

Word	POS	Additional POS features
smaller	ADJ	COMPARATIVE
entire	ADJ	ABSOLUTE ATTRIBUTIVE
fast	ADV	SUPERLATIVE
that	DET	CENTRAL DEMONSTRATIVE SG
all	DET	PREDETERMINER SG/PL QUANTIFIER
dog's	N	GENITIVE SG
furniture	N	NOMINATIVE SG NOINDEFDETERMINER
one-third	NUM	SG
she	PRON	PERSONAL FEMININE NOMINATIVE SG3
show	V	PRESENT -SG3 VFIN
show	N	NOMINATIVE SG
shown	PCP2	SVOO SVO SV
occurred	PCP2	SV
occurred	V	PAST VFIN SV

Figure 5.11 Sample lexical entries from the ENGTWOL lexicon described in Voutilainen (1995) and Heikkilä (1995).

POS Taggers - Stochastic Tagger

- ➔ Another technique of tagging is Stochastic POS Tagging.
- ➔ Here the question that arises here is which model can be stochastic.
- ➔ The model that includes frequency or probability (statistics) can be called stochastic.
- ➔ Any number of different approaches to the problem of part-of-speech tagging can be referred to as stochastic taggers.
- ➔ Two Approaches:
  - ◆ ***Word Frequency Approach***
    - In this approach, the stochastic taggers disambiguate the words based on the probability that a word occurs with a particular tag.
    - We can also say that **the tag encountered most frequently with the word in the training set is the one assigned to an ambiguous instance of that word.**
    - The main issue with this approach is that it may yield an inadmissible sequence of tags.

POS Taggers - Stochastic Tagger

- Two Approaches:
  - ***Tag Sequence Probabilities***
  - It is another approach of stochastic tagging, where the tagger calculates the probability of a given sequence of tags occurring.
  - It is also called the n-gram approach.
  - It is called so because the best tag for a given word is determined by the probability at which it occurs with the n previous tags.

POS Taggers - Stochastic Tagger

- ➔ Two Approaches:
  - ◆ ***Properties of Stochastic POST Tagging***
    - Stochastic POS taggers possess the following properties −
    - This POS tagging is based on the probability of the tag occurring.
    - It requires training corpus
    - There would be no probability for the words that do not exist in the corpus.
    - It uses different testing corpus (other than training corpus).

POS Taggers - Transformation Based Tagger

- ➔ Transformation Based Tagging is also called as Brill Tagging
- ➔ This tagger is based on the concept of Transformation-Based Learning (TBL) approach
- ➔ TBL uses Supervised Learning
- ➔ There is assumption of pre-tagged training corpus
- ➔ It draws inspiration from both rule-based and stochastic taggers
  - ◆ Like rule-based taggers, TBL is based on rules that specify what tags should be assigned to what word
  - ◆ And like stochastic taggers, TBL is a machine learning technique, in which rules are automatically induced from data

POS Taggers - Transformation Based Tagger

- ➔ The TBL algorithm has a set of tagging rules.
- ➔ A corpus is first tagged using the broadest rule, that is, the one that applies to the most cases. Then a slightly more specific rule is chosen, which changes some of the original tags.
- ➔ Next an even narrower rule, which changes a smaller number of tags (some of which might be previously
- ➔ changed tags)

POS Taggers - Transformation Based Tagger

- Label the training set with most frequent tags.
  - **The can was rusted**

The	Can	Was	rusted
DET	MD	VBD	VBD

  - Add transformation rules to reduce training mistakes

The	Can	Was	rusted
DET	NN	VBD	VBN

- Similarly, rusted can be VBD and VBN
  - In grammar, VBN is preceded by VBD
  - And in corpus VBD-VBD pair is negligible

- '**Can**' can be Noun as well as Modal verb
- TBL uses ML as well as grammar rules
  - Modal verb is never preceded by the determiner
  - Also in corpus, possibility of 'determiner modal verb' pair is zero
- Hence '**can**' is replaced by **Noun** Tag

POS Taggers - Transformation Based Tagger

- ➔ Before the rules apply, the tagger labels every word with its most-likely tag.
- ➔ We get these most-likely tags from a tagged corpus. For example, in the Brown corpus, race is most likely to be a noun
  - ◆ P(NN|race)=0.98
  - ◆ P(VB|race)=0.2

Is	expected	to	race	tomorrow
VBZ	VBN	TO	NN	NN

race/NN is incorrect in above sentence

The	Race	for	outer	space
DET	NN	IN	JJ	NN

race/NN is correctly tagged in above sentence

POS Taggers - Transformation Based Tagger

- ➔ After selecting most likely tag. Brill's tagger, applies transformation rules
- ➔ Change NN to VB when previous tag is TO
- ➔ Hence the correction would be

Is	expected	to	race	tomorrow
VBZ	VBN	TO	VB	NN

POS Taggers - Transformation Based Tagger

#	Change tags From	To	Condition	Example
1	NN	VB	Previous tag is TO	to/TO race/NN → VB
2	VBP	VB	One of the previous 3 tags is MD	might/MD vanish/VBP → VB
3	NN	VB	One of the previous 2 tags is MD	might/MD not reply/NN → VB
4	VB	NN	One of the previous 2 tags is DT	
5	VBD	VBN	One of the previous 3 tags is VBZ	

Figure 5.22 The first 20 nonlexicalized transformations from Brill (1995).